# L04 Resistors set the current

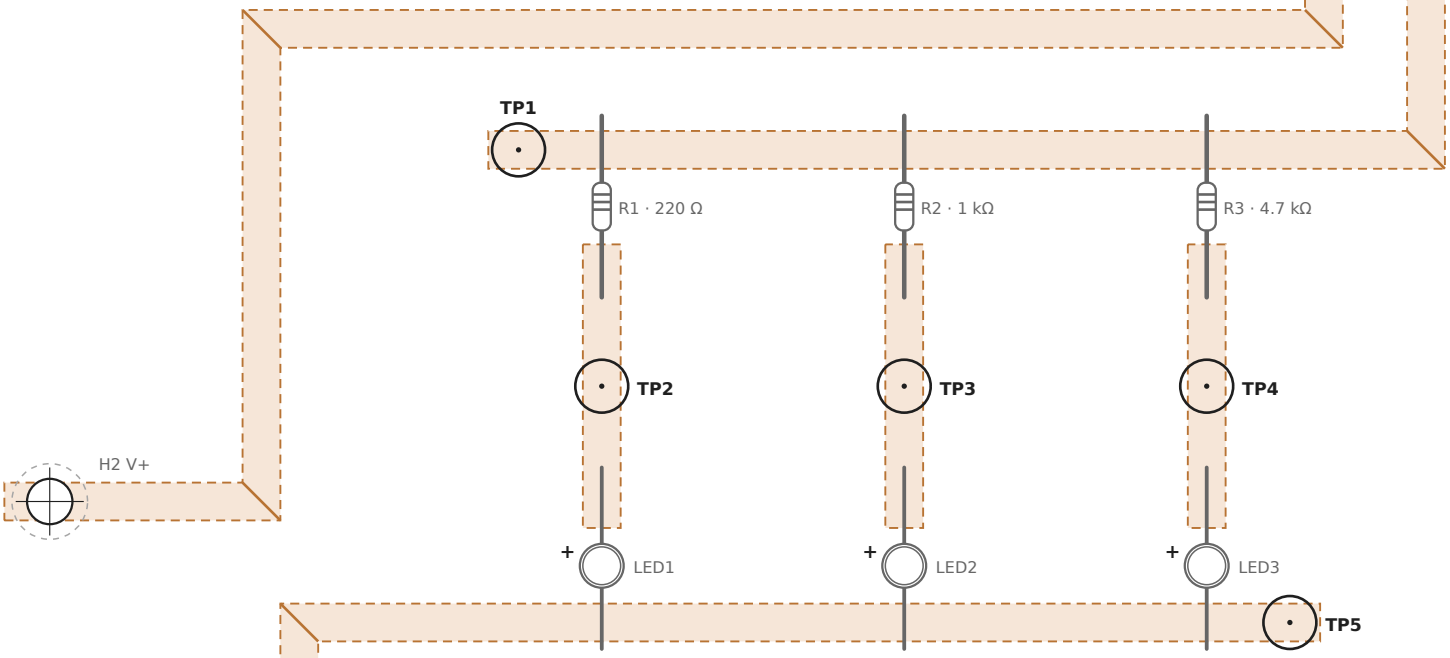
L04

Same LED, three resistors. Read the colours, then measure the current.

Draft: tagged steps may change

H1 BUS A · NC

S1
fold the corner down




H2 V+

H3 GND

## You need

- 3 red LEDs, 5 mm
- R1 220 Ω, R2 1 kΩ and R3 4.7 kΩ **[P]**
- 5 mm copper tape
- Power page P1 in the binder
- A multimeter for Measure

### Colour code

Band 1, band 2, then that many zeros.

| | | | |
|---|---|---|---|
| **0** | black | **5** | green |
| **1** | brown | **6** | blue |
| **2** | red | **7** | violet |
| **3** | orange | **8** | grey |
| **4** | yellow | **9** | white |

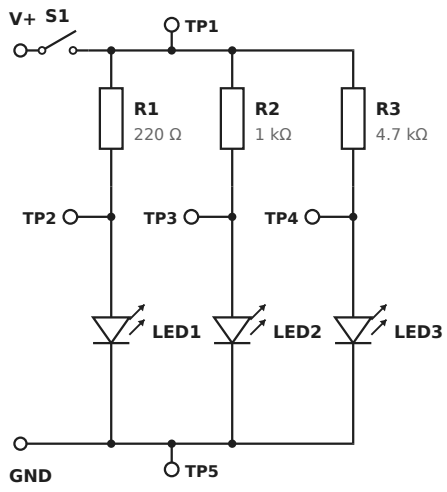
e.g. yellow-violet-red = 4, 7 + 00 = 4.7 kΩ

## Build it

1. Read each resistor's bands with the colour code below, and sort them into R1, R2 and R3.
2. Lay the six lanes, one piece of tape each. Fold at the creases; never cut.
3. Stick a 10 mm piece of copper tape on the corner flap, S1. **[T0]**
4. Punch the holes and set eyelets in H2 and H3. **[T0]**
5. Place R1, R2 and R3 across their gaps, either way round. Check each against its label.
6. Place LED1, LED2 and LED3 as printed: long leg on the +.
7. Hold each lead flat on its lane with a small patch of copper tape. **[T0]**
8. Bind this page with P1, then fold the corner down to switch on.

**[T0]** and **[P]** mark choices that may change: **[T0]** after the contact tests, **[P]** once the power supply is settled.

### Schematic




V+ and GND come from P1 through the binder.

H4 BUS B · NC

**Predict** Rank the LEDs from brightest to dimmest.

**Brightest** _____ _____ _____ **Dimmest**

Which resistor do you think lets the most current through, and why?

**Measure** Hold the corner down. Red probe on TP1, black on TP2–TP4.

| Across | Volts | mA |
|---|---|---|
| R1 220 Ω | | |
| R2 1 kΩ | | |
| R3 4.7 kΩ | | |

**mA = volts ÷ kΩ**

220 Ω is 0.22 kΩ

**Explain** What happens to the current when the resistance goes up?

Write it as a rule. Did the voltage across each resistor change much?

Print at 100 %: this bar should measure 100 mm.